INCH-POUND

MIL-PRF-49082E (CR)
4 March 1999
SUPERSEDING
MIL-V-49082D(CR)
22 May 1992

PERFORMANCE SPECIFICATION
VIEWER, DRIVER'S, NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Viewer, Driver's Night Vision AN/VVS-2(V).

1.2 Classification. Driver's Viewers covered by this specification must be of the following types as specified:

Type I	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)lA (M60 application).
Type II	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)2A (XMl application).
Type III	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)3 (FVS application).
Type IV	-	Viewer, Driver’s, Night Vision, AN/VVS-2(V)4

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.2 Government Documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
MILITARY

MIL-PRF-49040	-	Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter, MX-9644/UV
MIL-PRF-49147	-	Lens Assembly, Eyepiece, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-PRF-49148	-	Lens Assembly, Objective, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-PRF-49149	-	Electronic Control Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-PRF-49150	-	Entrance Housing Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-PRF-49151	-	Power Adapter Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2.2 Reference documents, drawings and publications. The following other documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS
USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771480-1	-	Viewer, Driver's Night Vision AN/VVS-2(V)lA
SM-D-771480-2	-	Viewer, Driver's Night Vision AN/VVS-2(V)2A
SM-D-771480-3	-	Viewer, Driver's Night Vision AN/VVS-2(V)3
5005820-1	-	Viewer, Driver’s Night Vision AN/VVS-2(V)4

(Copies of specifications, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Viewer, Driver's Night Vision, AN/VVS-2(V), herein referred to as the viewer, is a biocular display viewer used for closed hatch driving of combat vehicles. The viewer consists of an objective lens assembly, eyepiece lens assembly, entrance housing assembly, main viewer housing assembly, which includes an electronic control assembly and an electric power adapter, and the tube and magnifier assembly.

3.2 Construction. The viewer shall be constructed to assure compliance with the contract configuration baseline and in accordance with the following drawings as applicable:

TYPE I	-	SM-D-771480-1
TYPE II	-	SM-D-771480-2
TYPE III	-	SM-D-771480-3
TYPE IV	-	5005820-1

3.3 First article testing (FAT). When specified in the contract (see 6.2.e) the contractor shall furnish unit(s) for FAT inspection and approval (see 4.2).

3.4 Materials. The contractor shall select the parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Major components. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.d).

3.6 Performance characteristics.

3.6.1 ON/OFF/Gain switch. The switch shall turn the viewer ON and OFF, and vary the brightness of the display. Verification shall be performed using the power adapter powered by a 28 ± 2Vdc source and also using the BA5567/U battery or equivalent.

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3.6.2 Viewer bearing rotation. The viewer shall rotate 45 degrees in either direction from the center detent position for Type I and Type IV, and 30 degrees in either direction from the center detent position for Type II and Type III. Torque required shall be between 3 and 6 foot pounds, excluding the detent position.

3.6.3 Electronic control. When installed in the viewer the electronic control shall turn off the viewer when the ambient light level exceeds 4 foot lamberts. Viewer operation shall be restored within 5 seconds after the brightness of the target area is reduced below the cutoff value of the photodetector.

3.6.4 Field of view. The viewer's field of view shall have a minimum width of 44 degrees horizontal and 35 degrees vertical.

3.6.5 Resolution. The viewer shall have a minimum on-axis resolution of 0.8 line pairs per milliradian at optimum target illumination.

3.6.6 Magnification. The on-axis magnification of the viewer shall be 0.93 +0.09/-0.06 measured at 50 millimeter (mm) eye relief.

3.6.7 Linear distortion. The linear distortion of the viewer measured from an on-axis pupil at 50 mm eye relief shall not exceed 13 percent.

3.6.8 Objective focus. The objective shall be fixed focus for an object distance of 50, ± 5 feet.

3.6.9 Eyepiece focus. The eyepiece focus when measured from 50 mm eye relief shall be set at -2.0, +0.3/-0.1 diopters on axis.

3.7 Environmental characteristics.

3.7.1 Temperature Extremes. The viewer shall not be damaged nor suffer degradation of performance (see 3.13.1 and 3.13.5) after being operated at +52^{o}C or -51^{o}C or after storage at +68^{o}C or -51^{o}C. The viewer shall not be damaged nor suffer degradation of performance after being subjected to repetitive temperature changes between +23^{o}C and +68^{o}C in 5 minutes and between +23^{o}C and -51^{o}C in 5 minutes (see 3.13.1 and 3.13.5). Soak time at each temperature extreme shall be sufficient for the viewer to attain temperature stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.

3.7.2 Humidity. The viewer shall not be damaged nor suffer degradation of performance (see 3.13.1 and 3.13.5) after being subjected to relative humidities of greater than 90 percent and temperatures varying from +20^{o}C to +68^{o}C with exposure not less than 240 hours.

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3.7.3 Immersion. The viewer shall not be damaged (see 3.13.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The viewer shall be pre-conditioned by exposure to temperature of 45^{0}C ±3^{0}C for a period of 2 hours prior to being immersed in the fresh water at a temperature of 18^{0}C ±5^{0}C.

3.7.4 Vibration. The viewer shall withstand vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation without damage or degradation of performance (see 3.13.1 and 3.13.5).

a. Vibration shall be performed in 3 orthogonal axes. Vibration shall be a sine sweep from 5 to 500 Hz and back to 5 Hz within 15 minutes. Vibration shall be performed from 5.0 to 5.5 Hertz (Hz) at 1 inch displacement, 5.5 to 30 Hz. at 1.5g's peak, 30 to 50 Hz. at 1 inch displacement at 0.33 inch displacement and 50 to 500 at 4.2 g's peak. Duration shall be ten (10) sweeps for a total of 150 minutes.

b. Vibration shall be limited to one (1) sweep in the vertical axis. Vibration shall be a sine sweep from 5 to 500 Hz and back to 5 Hz within 15 minutes. Vibration shall be performed from 5.0 to 5.5 Hertz (Hz) at 1 inch displacement, 5.5 to 30 Hz. at 1.5g's peak, 30 to 50 Hz. at 1 inch displacement at 0.33 inch displacement and 50 to 500 at 4.2 g's peak.

3.7.5 Shock.

3.7.5.1 Basic shock. The viewer shall withstand a basic shock pulse of 40 g's (see 3.13.3) with a pulsewidth of 18 milliseconds without damage or degradation of performance (see 3.13.1 and 3.13.5).

3.7.5.2 Weapon shock. The viewer shall withstand simulated weapon firing shock pulse of 100 g's (see 3.13.3) with a pulsewidth of 1.5 millisecond one half sine pulse without damage and degradation of performance (see 3.13.1 and 3.13.5).

3.7.6 Fungus. The viewer shall neither support fungus growth, be damaged nor suffer degradation of performance (see 3.13.1 and 3.13.5) by the presence of fungus spores or adjacent fungal growth after exposure to viable fungal spores for not less than 28 days with temperature and humidity cycled from 30 ±1^{0}C and 95 ±5 percent relative humidity for a period of 20 hours and 25 ±1^{0}C and 100 percent relative humidity for a period of 4 hours.

3.7.7 Salt fog. The viewer, with protective covers installed, shall not be damaged (see 3.13.1) after 48 hours minimum exposure to a salt fog atmosphere of 5 percent sodium chloride containing, on the dry basis, not more than 0.1 percent sodium iodide and not more than 0.5 percent of total impurities at a temperature of 35 ±2^{0}C.

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3.7.8 Sand and Dust. The viewer, with protective covers installed, shall not be damaged nor suffer degradation of performance (see and 3.13.1 and 3.13.5) by exposure of the portion of the viewer external to the vehicle to the environment of sand and dust blowing at 35 knots. Dust used in this test shall be 97 - 99 percent by weight SiO_2 of angular structure with a size distribution as determined by weight, such that the following is met when using the U.S. Standard Sieve Series:

a. 100 percent of this dust shall pass through a 100 mesh screen.
b. 98 percent of this dust shall pass through a 140 mesh screen.
c. 90 percent of this dust shall pass through a 200 mesh screen.
d. 75 percent of this dust shall pass through a 325 mesh screen.

NOTE: 140 mesh silica flour as produced by the Ottwa Silica Company, Ottwa Illinois, or equal is satisfactory for use in the performance of this test.

Test conditions shall be such that dust is introduced at a rate of 0.3 ±0.2 grams per cubic foot into an air velocity of 1750 ±250 feet per minute at a temperature of 23 ±2°C and a relative humidity of less than 22 percent for a period of 6 hours. At the completion of the 6 hours period, the dust feed shall be stopped, the air velocity reduced to 300 ±200 feet per minute and the air temperature raised to 63 ±2°C for a period of 16 hours. At the completion of the 16 hour period, the air temperature shall be maintained at 63 ±2°C, the dust feed shall be returned to 0.3 ±0.2 grams per cubic foot, and the air velocity shall be returned to 1750 ±250 feet per minute for a period of 6 hours. A the completion of the 6 hours, allow the viewer to return to standard ambient conditions. Remove accumulated dust from the viewer by brushing, wiping or shaking taking care not to introduce additional dust into the viewer. Dust shall not be removed by air blast or vacuuming. After removal of accumulated dust, the controls of the viewer shall be operated and operation verified.

3.8 Reliability. The mean time to failure (MTTF) of the viewer shall be 1000 hours of operation minimum in a tracked vehicle simulated field environment.

3.9 Interchangeability. The entrance housing assembly, the objective lens assembly and the eyepiece lens assembly shall be physically interchangeable between viewer assemblies without degradation of the resolution (see 3.6.5), objective focus (see 3.6.8) and eyepiece focus (see 3.6.9) requirements. These interchanges shall be made without any modification or adjustment beyond that necessary during replacement of any of the parts or assemblies involved.

3.10 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on applicable drawings unless otherwise specified in the contract.

3.11 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract.

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3.12 Workmanship. Standards of workmanship shall be such that the viewer will meet all requirements of this specification and any referenced specifications or drawings

3.13 Technical interpretations. The following technical interpretations are, when referenced in sections 3 mandatory for this specification.

3.13.1 Damage. Damage is defined as the following:

a. Breakage, loosening, shift, dents, failure of finish, hardware, connection or component.
b. Condensation, fogging or moisture on any internal surface.
c. Failure to operate or operate properly.

3.13.2 Chargeable failure. A chargeable system failure during the reliability test of 3.8 is the inability to pass the tests of 3.6.1, 3.6.2, 3.6.3, or 3.6.5 (see 4.4).

3.13.3 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

3.13.4 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.13.5 Degradation of performance. Degradation of performance is defined as failure to operate in accordance with the requirements of this specification. For post-environmental test verification, degradation of performance shall be defined as failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, or 3.6.5 of this specification.

3.13.6 Environmental temperature. Environmental chamber temperatures shall be controlled to within $\pm 5^{0}$C of the specified temperature unless otherwise specified.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article testing (4.2).
b. Conformance inspection (4.3).

4.2 First article testing. Unless otherwise specified in the contract or purchase order, first article testing shall be performed by the contractor. Unless otherwise specified, the FAT testing shall consist of ten (10) viewers.

4.2.1 FAT unit tests. Each FAT viewer shall pass the requirements of Table I. Failure of any test shall constitute failure of that unit and may constitute failure of the FAT.

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4.2.2 FAT sample test. FAT sample testing shall be in accordance with Table II. Three (3) viewers shall be used in Subgroup 1, three (3) viewers shall be used in subgroup 2 and four (4) viewers shall be used in Subgroup 3. In Subgroup 2, one (1) each viewer shall be used for salt fog, sand and dust, and fungus. FAT test shall be conducted in the order listed except subgroup 2, which may be conducted concurrently. Failure of any test shall constitute failure of that unit and may constitute failure of the FAT.

TABLE I. First article testing unit test.

Inspection	Requirement Paragraph	Subgroups 1	2	3
ON/OFF Gain Switch	3.6.1	X	X	X
Viewer bearing rotation	3.6.2	X	X	X
Electronic control	3.6.3	X	X	X
Field of View	3.6.4	X	X	X
Resolution	3.6.5	X	X	X
Magnification	3.6.6	X	X	X
Linear distortion	3.6.7	X	X	X
Objective focus	3.6.8	X	X	X
Eyepiece focus	3.6.9	X	X	X

TABLE II. First article testing sample test.

Inspection	Requirement Paragraph	Subgroups 1	2	3
Temperature Extremes	3.7.1	X		
Humidity	3.7.2	X		
Vibration	3.7.4	X		
Shock	3.7.5	X		
Immersion	3.7.3	X		
Fungus	3.7.6		X	
Salt Fog	3.7.7		X	
Sand and Dust	3.7.8		X	
Reliability	3.8			X

4.2.3 Disposition of FAT samples. After successful completion of the FAT, the FAT units shall be refurbished and delivered in accordance with the contract.

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order.

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4.4 Operational tests. Viewer failures attributable to tube and magnifier assembly failure shall not be counted against the viewer. Failed tube and magnifier assemblies shall be replaced and the test resumed at the point of failure occurrence. Only salt fog and sand and dust shall be performed with the protective covers installed. The operational tests required during environmental testing shall consist of turn on of the viewer and observance of the green glow at the eyepiece. Either an external power supply furnishing 28, ± 2Vdc to the power adapter assembly or a battery may be used for all tests. Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.13.5). Temperature chambers shall maintain specified temperatures within ± 2.0°C.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order(see 6.2 g). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data is retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The viewer is intended for closed hatch driving of combat vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. A prohibition against delivering viewers with batteries installed.
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
d. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
e. Statement specifying that FAT testing is, or is not, required.
f. Statement identifying that the contractor must submit his proposal for Environmental Stress Screening (ESS).
g. Packaging requirements.

6.3 Definitions. see 3.13.

6.4 Subject term keyword listing.

Electronic control

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Eyepiece focus
Field of view
Linear distortion
Magnification
Objective focus
ON/OFF/Gain switch
Resolution
Viewer bearing rotation

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army-CR

Preparing Activity:
Army-CR
Project: 5855-0129

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49082E (CR)	**2. DOCUMENT DATE (YYMMDD)** 990304
3. DOCUMENT TITLE Viewer, Driver's, Night Vision, AN/VVS-2(V)		
4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)		
5. REASON FOR RECOMMENDATION		
6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)
8. PREPARING ACTIVITY		
a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129	
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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